4. It is a decentralised system and hence there is no single point of control or a single point of failure either.

HydraChain

This is an open source extension of the Ethereum blockchain platform that helps in developing and deploying different permissioned distributed ledgers. HydraChain is compatible with the Ethereum protocol and also provides the infrastructure to develop smart contracts in Python. It supports many tools and hence helps to reduce development time and improve debugging capabilities. It also provides a high level of customisation — different aspects of the system can be easily configured to achieve customer needs.

Features

1. It is highly compatible at an API and contract level. All the existing toolchains used to develop or deploy smart contracts can easily be used again.
2. HydraChain relies on registered and accountable validators that propose and validate the order of different transactions.
3. It supports Instant Finality.
4. HydraChain provides the infrastructure to develop various smart contracts in the high-level language, which reduces development times and improves debugging capabilities. This also increases the execution of native contracts.
5. Many aspects of HydraChain can be easily configured to match the customers' needs.
6. It supports easy deployment. A test network can be set up with almost zero configuration.

Advantages

1. Available as open source.
2. Different support plans and consulting options are provided as per customer requirements.
3. Easily scalable to support multiple linked chains in the same process.
4. High level of transaction privacy.

Corda

Corda is one of the preferred open source blockchain platforms for building and developing various permissioned distributed ledger systems. It was created by the R3 consortium, which comprises the largest banks and allows them to manage all the legal agreements between parties. It can be used by diverse businesses like financial institutions in order to keep a shared ledger of all the transactions, hence removing the need for all the involved parties to constantly keep a check on their books after they interact with each other. This is one of the primary problems for which Corda was developed. Similar to other blockchain platforms, R3 Corda also provides safe data storage and immutable records of data. One of the significant things about Corda is that it enables us to develop interoperable blockchain networks that can transact in strict privacy. At the moment, it is probably the only distributed ledger platform that has got a pluggable consensus.

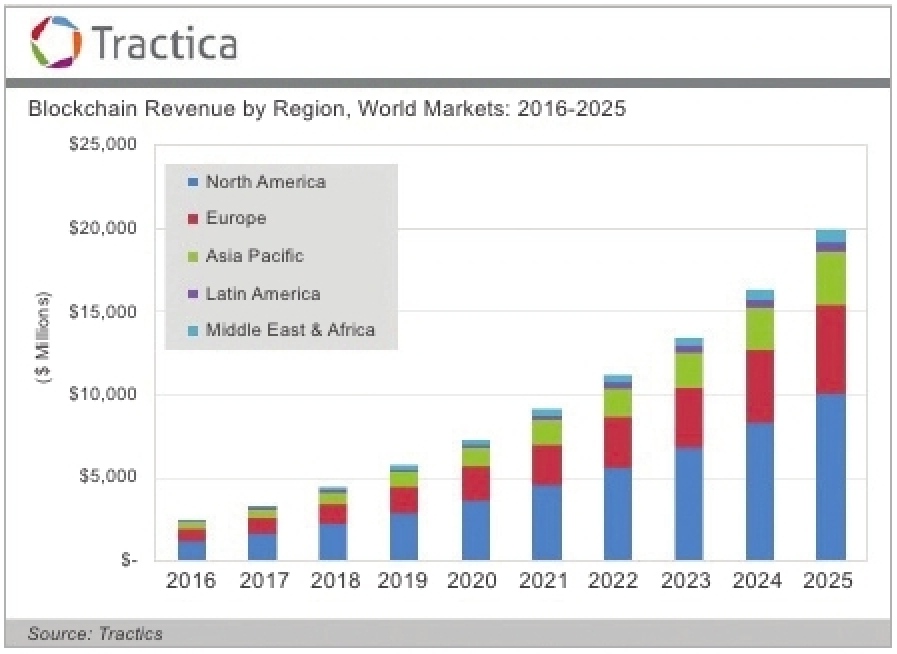

Figure 5: Estimation of revenue generation by blockchain technology across the globe (Image source: *googleimages.com*)

Features

1. The entire transaction history on the Corda ledger is completely encrypted and is shared only with the required.
2. All the different versions of Corda can easily interoperate with each other on the Corda network.
3. Corda is really agile. It is a flexible platform that can scale itself to meet all business needs.
4. It is an open source platform that can deliver a suitable blockchain for any business.
5. It is backed by a robust community of experts who are continuously working on its enhancements, functionalities, features and more.

Advantages

1. As Corda is available as open source, its source code is also available on GitHub under an Apache 2 licence.
2. A highly flexible and agile platform.
3. Easily scalable to meet customer needs.
4. Supported by an expert community of developers. END

References

[1] *http://www.wikipedia.org/*
[2] *https://blockchainhub.net/*
[3] *https://medium.com/*
[4] *https://www.coindesk.com/*

By: Vivek Ratan

The author has completed his B. Tech in electronics and instrumentation engineering. He currently works as a senior automation test engineer at Tata Technologies, Pune, and as a freelance educator at LearnerKul, Pune. He can be reached at *ratanvivek14@gmail.com*.